The Rank-21 Program: Reproduction, Soundness, Exhaustive Negatives, and a Certified-Search Architecture for $\langle 3,3,3\rangle$ over $\mathbb{F}_2$

Greg Sidebottom

August 31, 2026 (working note); postscript September 2, 2026

Abstract

We report on a program aimed at improving the lower bound for the rank of the 3×3 matrix multiplication tensor over $\mathbb{F}_2$, currently $20 \leq R_{\mathbb{F}_2}(\langle 3,3,3\rangle) \leq 23$ (Wang's automated lower bound; Laderman's scheme reduced mod 2). The contributions are: (i) an independent reproduction and strengthening of Wang's certified DP-over-restrictions bound, including budget-scaled lifts of his orbit lattice; (ii) a sound extension of his framework to *mixed-side* substitution, including an activity guard whose necessity we demonstrate by measurement, and a family of exhaustive negative results showing that substitution with flattening-class leaves cannot exceed the lattice values; (iii) an exact determination of our substitution game's value (18) under successively stronger rule sets, with machine-checked certificates; (iv) a certified-search (CDCL(T)-style) architecture over scheme space whose propagator and symmetry components each deliver their theoretical strength (measured ablations of 746× and 632,384×), reaching an independent, exhaustively-searched $R_{\mathbb{F}_2} > 15$ in 3.25 hours (root set completed in the postscript); and (v) an empirical structure theorem for the adversary's dangerous moves in the substitution game — *the Rank Law* — identifying them with rank-one covectors, which yields a concrete, well-posed lemma — proved by orbit exhaustion in the postscript, together with its level-2 form. We close with a convergence thesis: three structurally different attacks terminate at the same missing object, a cheap rank lower-bound oracle of strength 15–19 for arbitrary tensors, and we locate the genuine frontier at scheme size 20, where every known bound is structurally one short of what novelty requires.

1 Setting and known bounds

Let $\mathcal{T} = \langle 3,3,3\rangle \in \mathbb{F}_2^9 \otimes \mathbb{F}_2^9 \otimes \mathbb{F}_2^9$ denote the matrix multiplication tensor in the trace convention $\mathcal{T} = \sum_{i,j,k} x_{ij} \otimes y_{jk} \otimes z_{ki}$, a 729-bit object with 27 nonzero entries. Its rank $R_{\mathbb{F}_2}(\mathcal{T})$ is the minimum number of products $\alpha \otimes \beta \otimes \gamma$ (rank-one tensors) whose XOR is $\mathcal{T}$; schemes of r products are exactly solutions of the Brent equations with r multiplications. Known bounds: $R_{\mathbb{F}_2}(\mathcal{T}) \geq 20$ (Wang, arXiv:2603.07280, with a machine-verifiable certificate of ∼32 MiB), and $R_{\mathbb{F}_2}(\mathcal{T}) \leq 23$ (Laderman's scheme reduced mod 2; independently, our 55-addition rank-23 scheme). Bläser's ≥ 19 holds over every field. The program's goal is 21.

Throughout, *flattening* bounds are matrix ranks of the three unfoldings (≤ 9 here); K_p denotes the Koszul flattening with wedge parameter p (division bound $\lceil \mathrm{rank} / \binom{8}{p} \rceil$); and the *substitution lemma* is used in the form: for an active direction (one the tensor depends on), $R(T) \geq 1 + \min_\lambda R(T|_\lambda)$ where λ ranges over all foldings of that direction.

A structural datum that organizes everything below: rank methods — bounds expressible as matrix ranks of linearly-constructed flattenings, including Koszul, Young and Strassen's equations — are barrier-limited near $2n^2 = 18$ for matmul-type tensors. Our measurements respect this barrier with uncanny precision: flattenings reach 9, Strassen's commutator bound 12–13, Koszul 14–16, our exhaustive game with such leaves exactly 18. Wang's 20 lives beyond the barrier because substitution-DP is *recursive amplification*, not a rank method. This is the frame in which "new leaf mathematics" must be read (§6, §8).

2 Reproducing and strengthening the lattice DP

We reproduced Wang's result locally from his repository (verifier included), then strengthened his certificate by re-running his backtracking technique at 30× the shipped step budget over the whole restriction lattice. The wave lifted every layer — e.g. the fourteen codimension-2 orbits from $\{17{\times}1, 18{\times}13\}$ to $\{18{\times}6, 19{\times}8\}$ — but the root held at 20: a diminishing-returns profile, quantified. (A root of 21 requires, among more, all fourteen codim-2 orbits at ≥ 19; six remain at 18 and resisted 3×10^9 steps each.) Two engineering contributions accompany the reproduction: a diagnosis and fix of an allocator-retention memory ratchet in long runs (inline DFS-cache keys; heap pressure relief), and per-orbit byte-level certificate merging. All lifted certificates re-verify under his verifier.

3 Mixed-side substitution: soundness and negatives

Wang's framework restricts only the A-side. Motivated by our game measurements (A-only value 16 vs. mixed value 18; §4), we implemented the *fifth technique*: chains mixing A- and B-side substitutions inside his backtracking DFS, with a backward-compatible proof format (per-record leaf kinds, per-node side bits) and a mirrored verifier.

The activity guard. The naive mixed-leaf bound (chain length plus flattening of the doubly-restricted tensor) is *unsound*: pure-A chains get activity for free (A-flattening injectivity survives A-restriction), but under double-sided restriction a substitution can kill a direction the tensor no longer depends on, inflating the bound by +1 per inactive step. The sound credit is rank-based: $k_B = \dim B - \dim(W + D)$, where W is the kernel of the B-rows and D the dead B-subspace of the A-restricted tensor. We validated the formula on engineered deactivations (independent Python implementation and a C++ probe), then measured it in production: per-orbit docked credit ranged from 0 to 8.6×10^9 — the guard is not vacuous, and the unguarded rule's 99.99% "rescue rates" were partly phantom credits.

Exhaustive negatives. With soundness in place, equal-budget experiments returned a clean null: guarded mixed play lifted no codim-2 orbit at 3×10^8 steps (nor did A-only). Escalating down-lattice where exhaustion is affordable produced eight decisive verdicts, *all* FAILS: at codim-4 and codim-3, mixed substitution with table leaves cannot lift Wang's values by one, exhaustively. A three-sided variant (C-moves) inflates the game ($\geq 6.6\times$ nodes on refutations that were 16–40 minutes under two sides) with no observed proving power. The binding constraint is identified and measured: once play leaves the pure-A lattice, every available leaf collapses to flattening class ($\approx$14–16), and the adversary grinds targets down against them.

4 The exhaustive substitution game

Independently of Wang's framework, we built an exact game engine over (U_A, U_B, U_C) restriction states with sandwich-group canonicalization, stabilizer-reduced adversary fanouts, and memoized decision ladders. Its verdicts, each with machine-checked certificates replayed by an independent Python checker (including isomorphism edges): game value 17, then 18 under support/spanning refinements and rank-profile case splits; three exhaustive FAILS at 19 under every rule set tried. A methodological lesson is recorded with the mathematics: an early version "proved" $R_{\mathbb{F}_2}(\langle 2,2,2\rangle) \geq 12$ (true value 7) because kill moves scored on the zero tensor *and the checker re-encoded the same rule.* Certificate checkers validate transcription, not semantics; every new proof rule since has been gated on known exact values first.

5 Certified scheme search (CDCL(T) over products)

The constructive dual: prove $R_{\mathbb{F}_2} > r$ by exhausting r-product schemes, with theory propagators supplying the reasoning that resolution lacks. Decisions are whole products; the residual (tensor XOR chosen products) carries native parity; propagators prune by residual rank bounds; symmetry enters as constructive orbit enumeration.

Gates at 2×2. The stage-1 prototype reproduces both certified values: $r = 6$ UNSAT (matching our hydra_satsuma DRAT-verified result) and $r = 7$ SAT with an independently verified Strassen-class witness. The measured ablation: one ply of sound substitution reasoning cut the $r = 6$ tree from 3.22×10^9 to 4.3×10^6 nodes (746×).

Scaling to 3×3. Lazy enumeration over the $511^3 \approx 1.3 \times 10^8$ products; triple-layout residuals making every flattening nine contiguous 81-bit extracts; constructive first-product canonicalization under the sandwich group (133,432,831 → 211 orbit representatives; the level-1 frontier *is* the orbit count); level-2 stabilizer minimality (median 16×); a Strassen-commutator propagator (strength 12–13 at 10–30 μs, validated by a constructed-rank gate: tensors built as XOR of k products have rank $\leq k$ by construction); Koszul at the shallow levels. The ladder:

rung	verdict	nodes	wall
$r = 9$	UNSAT (flatten)	783,363 → 211	0.1 s
$r = 10$	UNSAT (probe)	211	0.1 s
$r = 11$–14	UNSAT (Koszul at level 1)	211 each	0.1 s each
$r = 15$	UNSAT (killer-directed deep probe)	211	3.25 h
$r = 16$	out of reach	—	≥ 70 core-h on one root, unresolved

The $r = 15$ rung is the architecture's current independent reach, $R_{\mathbb{F}_2} > 15$: two rungs past the Koszul ceiling, taken by *lazy adversary-directed recursion* — at each node a cheap-floor chain (flatten → Strassen → Koszul) settles most folds, and only the *killers* (Koszul-dropping folds) recurse, with targets falling into Strassen range within two levels. Fixed-sweep plies, by contrast, were measured intractable (16.7 hours resolving fewer than 8 of 211 roots). Engineering lessons are recorded honestly: Method-of-Four-Russians elimination, equality-gated, was *benched out* (dense wins do not transfer to the sparse wedge shape); a leading-bit rewrite of the 9×81 rank kernel gave a universal 2.1×; five unsound artifacts across the program (an α-canonical cut, a λ=0-only probe, side-slot pollution, activity-free mixed leaves, a row-reversed matrix inverse) were all caught by known-value or constructed-value gates before contaminating any claim.

6 The Rank Law

Instrumenting the $r = 15$ probes produced a dataset of every Koszul-dropping fold at the 211 roots (10,338 killers; the probe's pivot is always the cell $(2, 2)$; every drop is $14 \to 13$ exactly). Writing the folding covector as $\varphi = e_{(2,2)} + \lambda$ and viewing it as a 3×3 matrix over $\mathbb{F}_2$:

Observation 1 (Rank Law; 211 roots × 3 sides = 633 pairs)**.**

1. $\operatorname{rank}(\varphi) = 1 \implies$ *killer, with zero exceptions. These are the* $4 \times 4 = 16$ *rectangle indicators through the pivot cell — exactly the observed sixteen killers per pair.*
2. $\operatorname{rank}(\varphi) = 3 \implies$ *non-killer, with zero exceptions.*
3. $\operatorname{rank}(\varphi) = 2$*: non-killer at* 595 *of* 633 *pairs; killer only at* 38 *exceptional pairs concentrated on* 30 *roots (210 extra folds).*

The reading: the adversary's dangerous moves are *product-shaped* — rank-one covectors aligned with the tensor's own rank-one structure, the same objects our game's rank-profile splits singled out. The law immediately poses a well-defined lemma:

Conjecture 1 (case-split lemma)**.** *On the level-1 residuals (matmul minus one product, restricted as in the probe), folding by a covector of matrix rank* 3 *preserves* $\mathrm{K}_4 \geq 14$*.*

Proven — and extended across the rank-2 boundary, whose 38 exceptions appear tied to low-rank structure of the root representatives — the lemma removes 168–240 of every 256 probe evaluations, repricing the $r = 16$ rung from ≥ 1.5 core-years to days. Independent of that payoff, the law is the first structural characterization of the substitution game's critical directions that we are aware of, and the killer set's algebraic description (points of an affine determinantal pencil over $\mathbb{F}_2^8$) suggests it is provable. *(It is: see Section 7, where the lemma and its level-2 form are proved by orbit exhaustion, and where the repricing estimate above is corrected by measurement.)*

7 Postscript (September 2, 2026): the lemma proved, a symmetry defect repaired, and the $r = 16$ wall quantified

7.1 The case-split lemma, by orbit exhaustion

Fix a fold vector $v \in A = M_3(\mathbb{F}_2)$ (the covector of Section 6 is $\operatorname{tr}(\cdot\, v)$) and a rank-one $m = \alpha \otimes \beta \otimes \gamma$. The Koszul data of $(\mathcal{T} + m)|_{\ker v}$ is a sandwich invariant of the pair (m, v), and the sandwich group is transitive on the rank classes of v; so it suffices to fix v at a class representative and to enumerate the orbits of m under $\operatorname{Stab}(v) \times GL_3$. We enumerate them exactly by a staged lexicographically-minimal canonical form and require the orbit sizes to sum to $511^3 = 133{,}432{,}831$; every orbit is then evaluated.

Theorem 1 (case-split lemma, level 1)**.** *For every rank-one* m *over* $\mathbb{F}_2$ *and every* v *of matrix rank* 3*, on any side,* $\mathrm{K}_{\leq 4}\left((\mathcal{T} + m)|_{\ker v}\right) = 14$*. The side-A,* $p = 4$ *wedge rank never drops below*

912 *(threshold* 911*); over the* 7,029 *orbits the drop is* 0 *(6,505 orbits,* 122,193,407 *products) or exactly* 6 *(*524 *orbits,* 11,239,424 *products), each* 6 *split* 3+3 *across the two honest* 8*-dimensional wedges* $p = 3, 4$ *on* A/v *(both of rank* 459*).*

Theorem 2 (rank-2 boundary and rank-1 inheritance)**.** *For v of rank* 2 *(*14,070 *orbits) the fold kills on exactly* 108 *orbits (*774,144 *products,* 0.58%*), each dropping the wedge by exactly* $12 = 6 + 6$ *to Koszul* 13*; the boundary is a finite table. For v of rank* 1 *(*3,556 *orbits) the base value is* 900 *and every product is a killer; the perturbation can raise the wedge rank by* 6*, and the* 6*-quantization breaks (*$+22 = 10 + 12$*) only where it no longer matters.*

Gates: the encoded action fixes $\mathcal{T}$ term-by-term (random elements); random (m, v) transports agree; every (root, side, λ) fold of the $r = 15$ probes is transported to its representative and compared with the direct computation (162,048 triples, zero mismatches), and the predicted killer set of the Rank Law's roots reproduces the 10,338-line dump byte for byte. The necessary conditions for the 108 rank-2 killers are $\mathrm{tr}(\alpha\beta\gamma) = 1$, $\mathrm{tr}(v\beta\gamma) = 0$, $\operatorname{rank}\alpha \leq 2$, $\alpha \neq v$; no combination of up to three matrix-rank/trace invariants is a pure classifier, so the structural (isotypic) proof stays open — the exhaustion is the proof.

Theorem 3 (level 2)**.** *Two successive side-A folds quotient by a* 2*-dimensional* $U \subset A$*. The* 43,435 *such U fall into* 17 *pencil classes. Every class whose pencil contains a rank-one matrix, and the* $(2, 2, 2)$ *pencil with stabilizer of order* 288*, has* $\mathrm{K}(\mathcal{T}|_{\ker U}) \in \{12, 13\}$ *and admits no leaf for any m (the perturbation raises the wedge by at most* 12*, short of* 911*). The remaining eight classes keep Koszul* 14 *(wedge* 918*, the level-1 value) on* 96.65–99.90% *of products, the killers dropping by exactly* 12 *or* 18*. Over all* (U, m) *pairs the level-2 leaf fraction is* 71.92% *(*1.65 *million orbits evaluated;* 300 *random transports agree).*

7.2 A symmetry defect and its repair

The invariance gate of the lemma tool failed on first run: the scheme search's C-side action, $\gamma \mapsto P^{-1}\gamma R^{-1}$, had been ported verbatim from the game engine, which indexes C as (i, k); the search indexes C as (k, i), where the same group element acts as $R^{-T}\gamma P^{-T}$. The ported action is *not* a symmetry of $\mathcal{T}$, so the "211 first-product orbit representatives" of Section 5 were representatives of a fake partition: an audit under the correct action shows they meet 159 of the 211 true orbits, leaving 52 never probed. (The α and β stages were correct; the counts coincide because the defect is a bijection of the orbit set.) Repair: $r = 14$ re-exhausts on the corrected representatives in 0.7 s; the killer-directed deep probe proves rank ≥ 15 on all 52 missed orbits in 394 s. The $r = 15$ row of the ladder now rests on a complete, gate-verified representative set. The Rank Law's statements concern the dumped residuals themselves and stand; the lemma above quantifies over all products. The lesson belongs with the method acknowledgment: an orbit reduction must be gated on the object it claims to preserve, not on a comment saying it was verified elsewhere.

7.3 The $r = 16$ wall, quantified

With the lemma in hand we rebuilt the probe for the rung: a work-queue implementation that holds all twelve cores (the fanned and root-parallel modes tail to eight and one), removal of *no-op folds* (a fold into a zero slice is a relabeling, not a quotient; a minimal decomposition projects off any zero slice, so half of level-2 and three quarters of level-3 evaluations were wasted), fold-orbit pruning under the root's stabilizer (one representative per orbit proves the whole conjunction, since the rank bound is an isomorphism invariant), and a transposition table keyed by the reduced bases of the folded subspaces (an exact key: the residual is the projection along U onto the non-pivot coordinates, unique given U). Verdicts were gated against the serial probe at every step. Cumulatively, the $r = 15$ proof of one root went from 65,792 to 1,473 tasks, and the hardest root from 42.8 to 13.3 seconds. The $r = 16$ rung on root 0 nevertheless stayed unresolved at every attempt (30 minutes on 12 cores each; the August 31 attempt had reached 70 core-hours). The per-depth profile explains why (nodes evaluated after 29 minutes, memo run):

depth (target)	4 (12)	5 (11)	6 (10)	7 (9)	8 (8)
nodes evaluated	2,674	87,050	199,245	846,677	2,825,251
Koszul leaves	14	78,414	143,622	413,355	2,564,972
genuine failures	0	0	0	0	181,398

Pure side-A chains bottom out at depth 7–8 on pencils all three of whose combinations have rank ≤ 7; the probe then switches sides, and the mixed keys (seven A-folds and one B-fold: $2^{14} \times 256$) are millions, so the deep layers are side-switch fan-outs, not revisits (memo hit rate 15–50%). Balanced side selection (fold the side with most active coordinates) is verdict-equivalent but loses at $r = 15$ by two orders of magnitude, because it keeps every level's fan at 256. In 29 minutes the search had completed 36 of the roughly 3,000 depth-3 nodes: the tree is $\sim 100\times$ the budget with every lever applied.

Two oracle routes for the small residuals were tested. Plain SAT (cadical on a Tseitin encoding of "rank $\leq r$") is dead: a 9×9 pencil of known rank 9 needs 3.5 s for the satisfiable side and exceeds 60 s to refute rank 8; random pencils time out at every $r \in [8, 11]$. An exact pencil oracle via the Kronecker form is feasible but the textbook count is wrong over $\mathbb{F}_2$: brute force ($\min_X \operatorname{rank} X + \operatorname{rank}(I{+}X) + \operatorname{rank}(A{+}X)$, $n \leq 4$) shows an irreducible cubic block contributes $+2$, not $+1$, as do $(x + 1)^4$ and irreducible quartics; block additivity held in every sample. Such an oracle would settle depth 7 exactly and remove the depth-8 layer — about 70% of the work, a $3\times$ — and would not reach the rung. The wall is depths 5–6: 3- and 4-slice residuals that must be proved to have rank 10–11, where Koszul caps at 13 but fails on the hard instances and no cheap exact method is in sight.

Standing verdict. The independent ladder rests at $R_{\mathbb{F}_2} > 15$ on a complete representative set. The certified-search route's next rung is not a search-engineering problem; it needs a rank oracle of strength 10–11 on 3–4-slice residuals over $\mathbb{F}_2$, the same missing object as Section 8 names, one size smaller.

8 Convergence, the frontier at $r = 20$, and outlook

Three structurally different programs — mixed-side substitution inside Wang's DP, the exhaustive game, and certified scheme search — terminate at the same gap: a cheap ($\lesssim$ ms) rank lower-bound oracle of strength 15–19 for *arbitrary* tensors (lattice-restricted tensors have Wang's table; residuals do not). The barrier section explains why the gap is stubborn: that strength range spans the rank-method ceiling, so the oracle must itself be amplification-shaped.

For the program's actual goal, the frontier is sharper than any oracle question. Subadditivity plus Wang's certificate gives $\operatorname{rank}(T \oplus m_1 \oplus \cdots \oplus m_k) \geq 20 - k$ for free, which trivializes every rung $r \leq 19$; proving $R_{\mathbb{F}_2} > 20$ requires certifying $21 - k$ on *scheme-consistent* prefixes — beating subadditivity by exactly one, which at $k = 0$ is the theorem itself. The $+1$ must come from knowledge about minimal decompositions of $\mathcal{T}$ specifically. Two routes: (a) prove and iterate the Rank Law (killer varieties as case-split rules — the game's critical directions may thin as prefixes accumulate); (b) decomposition-structure theorems in the tradition of de Groote's uniqueness and Burichenko's symmetries, mined from the known scheme corpora and imported as WLOG prunes. On the constructive side, the corridor experiment established that prefix descent has no usable guidance signal (flattening is capped and constant; true schemes' intermediates are low-rank, not low-popcount), so the rank-22 search correctly factors as SLS near-misses (native-ANF Brent engine) finished or refuted by the exact endgame engine, which refutes junk prefixes in milliseconds.

All measurements, certificates, logs and the killer dataset are in the repository; every number above is reproducible from committed artifacts.

—

Acknowledgment of method. The program's productivity traces to two disciplines applied without exception: gate every new rule on known exact values (five unsound artifacts caught), and distrust suspicious nulls until the treatment is proven applied (three vacuous experiments unmasked). The instruments built for soundness — the activity-deficit counters, the killer dump — produced the mathematical discoveries.